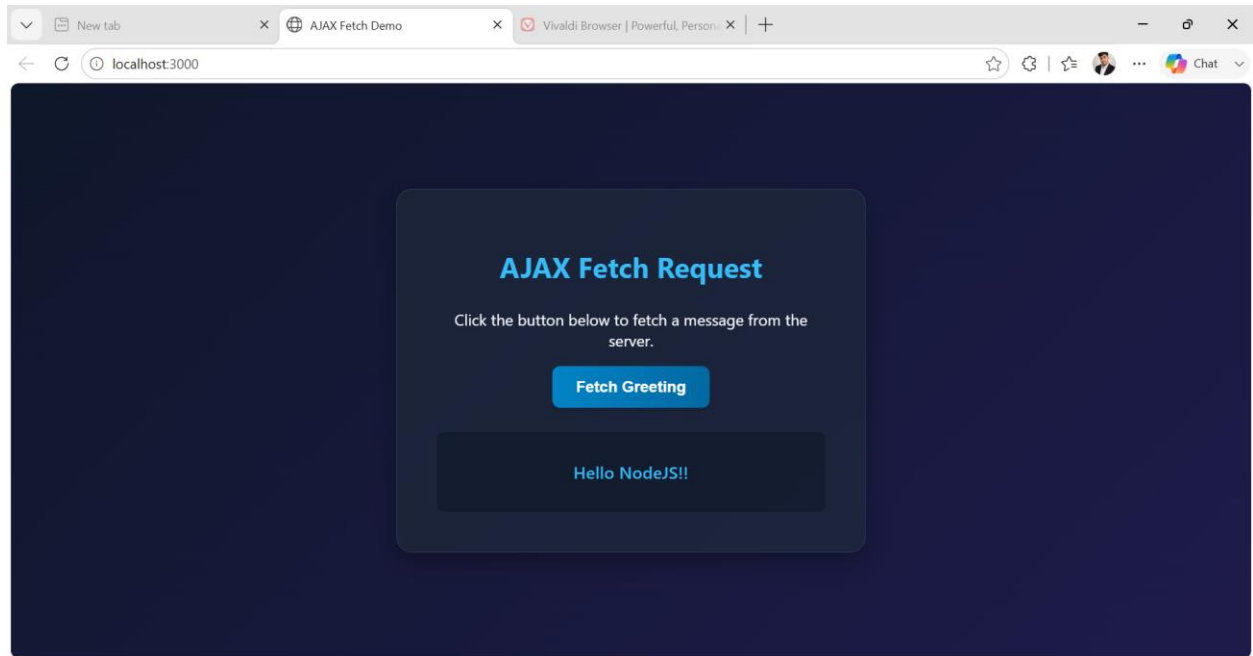
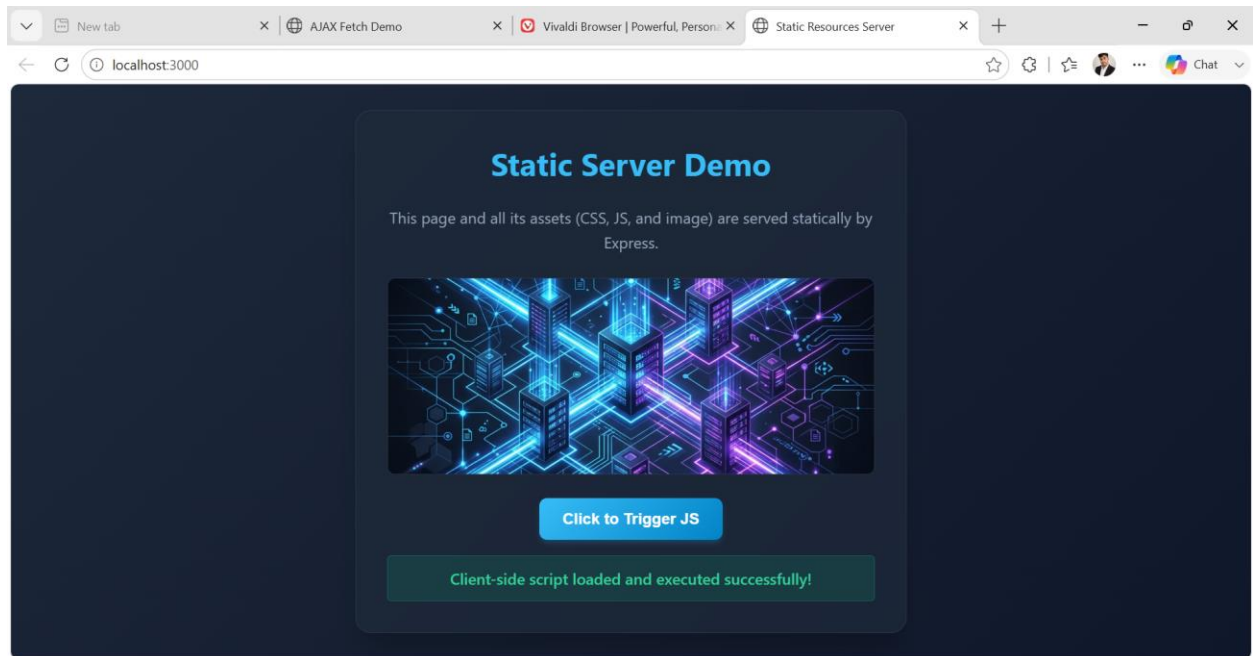


Question 1:



Question 2:



Question 3:

The screenshot shows the Antigravity IDE interface. The Explorer panel on the left displays a project structure for '33_AnshPatel_701_A1'. The main Terminal window shows the following commands and output:

```
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\02_static_server> npm start
Static server running at http://localhost:3000
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\02_static_server> cd ../03_chatbot_module
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\03_chatbot_module> node app.js
=====
Welcome to the College Helpdesk Chatbot CLI!
Ask me about admissions, fees, courses, exams, etc.
Type "exit" or "bye" to end the conversation.
=====

You: apply
Chatbot: For admissions, please visit our admissions portal at website/admissions or email admissions@college.edu. The deadline for the fall semester is August 31st.

You: tuition
Chatbot: Tuition fees vary by program. You can find the detailed fee structure at website/fees. Payments can be made via credit card, bank transfer, or at the student finance office.

You: bye
Chatbot: Thank you for visiting the College Helpdesk. Goodbye!
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\03_chatbot_module>
```

Question 4:

```
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\04_compress_folder> npm start

> 04_compress_folder@1.0.0 start
> node compress.js

Prepared source_folder with sample text files.
Successfully compressed directory to output.zip
Total size: 317 bytes
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\04_compress_folder>
```

Question 5:

```
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\05_extract_zip> npm start

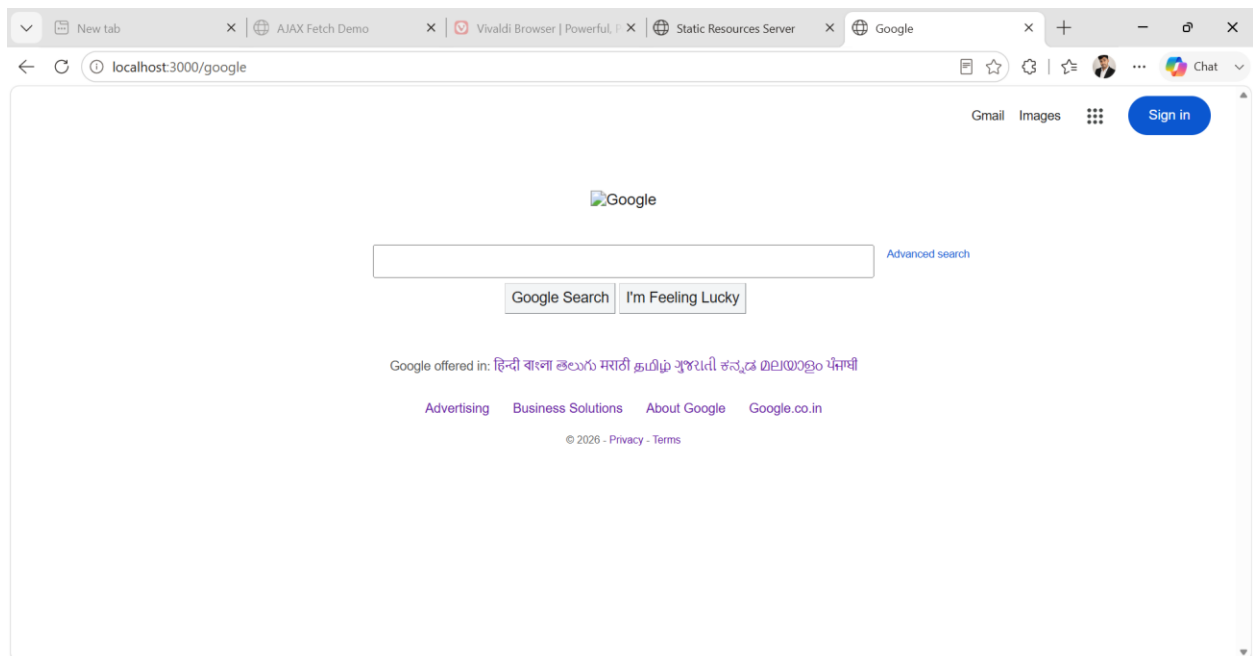
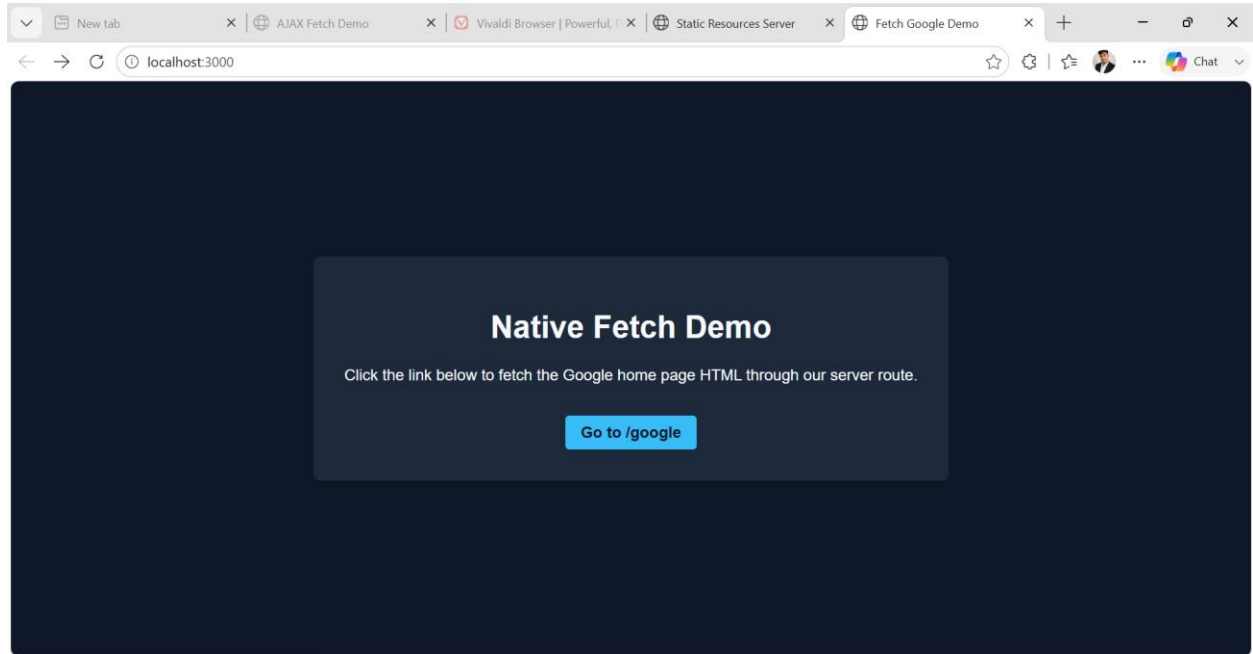
> 05_extract_zip@1.0.0 start
> node extract.js

Created sample zip file: sample.zip
Extracting sample.zip to extracted_folder...
Extraction completed successfully.
Extracted files: [ 'config.json', 'demo.txt' ]
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\05_extract_zip>
```

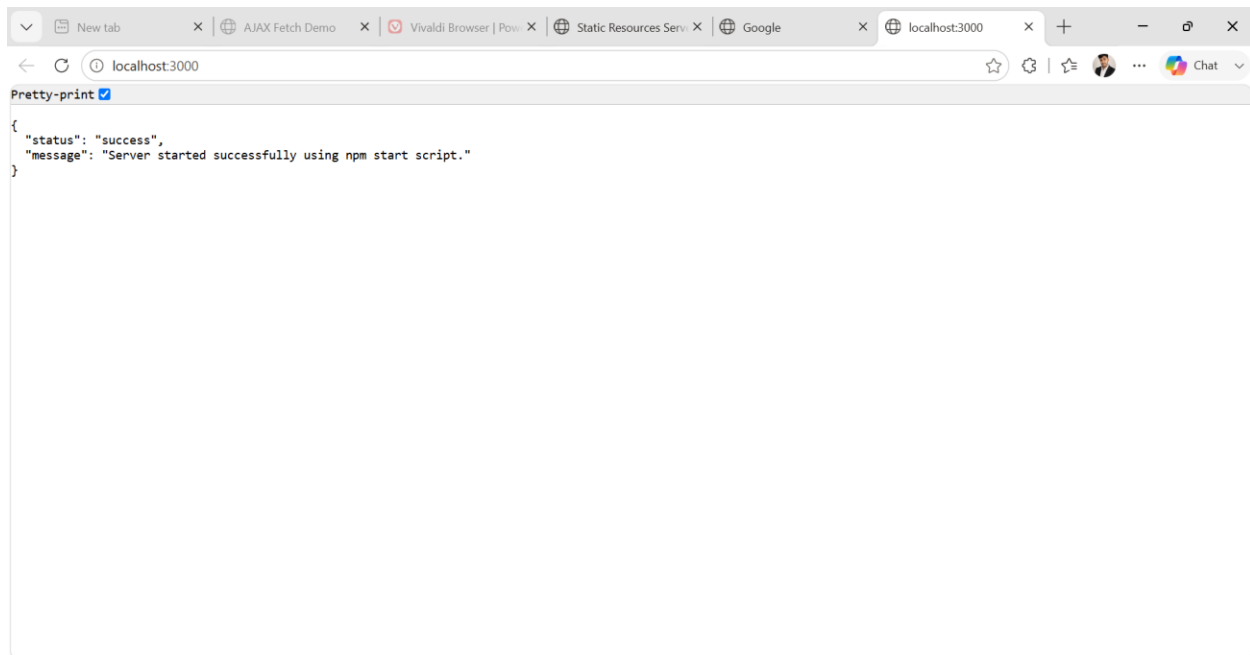
Question 6:

```
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\06_promisify_unlink> node unlink.js
Created temporary file: temp_to_delete.txt
Attempting to delete temp_to_delete.txt using promisified unlink...
File deleted successfully using async/await!
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\06_promisify_unlink>
```

Question 7:



Question 8:



```
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\08_npm_scripts> npm test

> 08_npm_scripts@1.0.0 test
> node test.js

Running test suite...
[PASS] Critical file found: server.js
[PASS] Critical file found: package.json
Test suite completed successfully.
```

```
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\08_npm_scripts> npm test

> 08_npm_scripts@1.0.0 test
> node test.js

Running test suite...
[PASS] Critical file found: server.js
[PASS] Critical file found: package.json
Test suite completed successfully.
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\08_npm_scripts> npm run build

> 08_npm_scripts@1.0.0 build
> node build.js

Running build script...
Build completed successfully. Created dist/bundle.js
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\08_npm_scripts> npm run lint

> 08_npm_scripts@1.0.0 lint
> node lint.js

Running lint script...
LINT WARNING: [lint.js:18] Use 'const' or 'let' instead of 'var'.
LINT WARNING: [lint.js:20] Use 'const' or 'let' instead of 'var'.
Lint completed with some warnings.
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\08_npm_scripts> npm run clean

> 08_npm_scripts@1.0.0 clean
> node clean.js

Running clean script...
Successfully cleaned the "dist" directory.
```

Question 9:

```
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\09_fs_operations> node fs_demo.js
=== Node.js File System (fs) Module Operations Demo ===

1. Creating directory: "demo_dir" using fs.mkdirSync...
Directory created successfully: true

2. Writing file: "test.txt" using fs.writeFileSync...
File written successfully.

3. Appending content to "test.txt" using fs.appendFileSync...
Content appended successfully.

4. Reading "test.txt" content using fs.readFileSync...
--- File Content Start ---
First line of content.
Second line of appended content.
--- File Content End ---

5. Fetching file metadata using fs.statSync...
- File Size: 56 bytes
- Is File: true
- Last Modified Time: Wed Jul 15 2026 21:07:46 GMT+0530 (India Standard Time)

6. Scanning directory contents using fs.readdirSync...
Files in "demo_dir": [ 'test.txt' ]

7. Renaming "test.txt" to "renamed_test.txt" using fs.renameSync...
Files in "demo_dir" after rename: [ 'renamed_test.txt' ]

=== fs operations demonstration completed ===
```

Question 10:

```
PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\10_global_objects> node globals.js student=Ansh c
lass=701 assignment=A1
=== Node.js Globals Demonstration ===

1. Accessing global variable (global.assignmentMetadata):
- Student: Ansh Patel
- Subject: 701
- Assignment: A1

2. File path globals:
- Current Directory (__dirname): C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\10_global_objects
- Current Filepath (__filename): C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\10_global_objects\g
lobals.js

3. Buffer usage (Encoding/Decoding a string):
- Original string: "Postgraduate Node.js Assignment 701"
- Raw Buffer representation: <Buffer 50 6f 73 74 67 72 61 64 75 61 74 65 20 4e 6f 64 65 2e 6a 73
20 41 73 73 69 67 6e 6d 65 6e 74 20 37 30 31>
- Base64 Encoded: UG9zdGdyYWR1YXRlIE5vZGUuanMgQXNzaWdubWVudCA3MDE=
- Decoded back from Buffer: Postgraduate Node.js Assignment 701

4. Process details and CLI arguments (process.argv):
- Node.js Version: v24.14.1
- OS Platform: win32
- Process Architecture: x64
- Process Uptime: 0.0274313 seconds
- Environment (process.env.NODE_ENV): Not Defined
- Command Line Arguments Passed:
  [0]: C:\Program Files\nodejs\node.exe
  [1]: C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\10_global_objects\globals.js
  [2]: student=Ansh
  [3]: class=701
  [4]: assignment=A1
```

Question 11:

```

PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\11_npm_package\consumer> npm start
> consumer@1.0.0 start
> node index.js

=== Custom NPM Package Consumer Test ===

1. Email Validations:
- "student@college.edu" is valid? true
- "invalid-email@" is valid? false
- "ansh.patel@gmail.com" is valid? true
- "test.com" is valid? false

2. Password Strength Validations:
- "12345" is strong? false
- "weakpassword" is strong? false
- "StrongPass123" is strong? true
- "ONLYCAPS1" is strong? false

3. Sanitizing inputs:
- Raw: "<script>alert('hack')</script> & safe text"
- Sanitized: "&lt;script&gt;alert('hack&quot;)&lt;/script&gt; &amp; safe text"

=== Consumer demonstration finished ===

```

Question 12:

```

PS C:\Users\Welcome\Desktop\33_AnshPatel_701_A1\12_promise_chaining> node chain.js
Initiating Promise Chain demonstration...

Starting Task 1: Fetching database record for student "Ansh Patel"...
Task 1 Completed: Found ID "70133" for Ansh Patel.

Starting Task 2: Fetching courses for Student ID 70133...
Task 2 Completed: Registered courses for Ansh Patel: Node.js Programming, Cloud Architectures, Web Security.

Starting Task 3: Fetching weekly schedule for course "Node.js Programming"...
Task 3 Completed: Schedule loaded successfully.
- Student: Ansh Patel
- Course: Node.js Programming
- Schedule: Mondays and Wednesdays at 10:00 AM in Room 402

=== Promise Chain demonstration completed ===

```